

Lectures On Class Field Theory

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1 Un peu d'histoire

This course will be about number fields (i.e. finite extensions $K|\mathbf{Q}$) and their absolute Galois groups $G_K = \text{Gal}(\overline{K}|K)$, where $\overline{K}$ is a fixed algebraic closure of K . Very little is known about the structure of such automorphism groups in general, and a large chunk of modern number theory/arithmetical geometry is devoted to studying them, e.g. via their representations. In some sense, class field theory consists in the study of its 1-dimensional representations; more precisely, it describes the abelianization $G_K^{\text{ab}} = G_K/[G_K, G_K]$ of the absolute Galois group, or equivalently its abelian quotients.

1.1 Reciprocity laws

Let p be a prime number, and let a be an integer coprime to p . The *Legendre symbol* $\left(\frac{a}{p}\right)$ is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square modulo } p, \\ -1 & \text{if } a \text{ is not a square modulo } p. \end{cases}$$

Euler proved that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \pmod{p}$, and **Gauss** proved the following law:

Theorem 1.1 (Quadratic reciprocity). Let $a \neq b$ be different odd primes. Then,

$$\left(\frac{a}{b}\right) \cdot \left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2} \frac{b-1}{2}}.$$

Moreover, the following subsidiary laws hold:

1. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}};$
2. $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$

These laws, together with its multiplicativity in the parameter a , make it possible to compute the Legendre symbol.

Another way of thinking about class field theory is as an attempt to define a “very general” reciprocity law, from which the quadratic reciprocity law and its higher degree analogues can be deduced. One such generalization is **Hilbert’s** reciprocity law: recall that $\mathbf{Q}$ embeds into the p -adic field $\mathbf{Q}_p$ for every prime p , and also into the real numbers $\mathbf{Q}_\infty := \mathbf{R}$. For every prime p (including $p = \infty$) and for all $a, b \in \mathbf{Q}^\times$, we define the *Hilbert symbol* $(a, b)_p$ as

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ has a solution in } \mathbf{Q}_p^2, \\ -1 & \text{otherwise.} \end{cases}$$

Recall that the norm $N : \mathbf{Q}_p(\sqrt{a}) \rightarrow \mathbf{Q}_p$ is defined as $N(x + y\sqrt{a}) = x^2 - ay^2$ if $a \notin \mathbf{Q}_p^{\times 2}$, and as $N = \text{id}$ otherwise. More generally, if $L|K$ is a finite Galois extension with group G , the norm $N_{L|K} : L \rightarrow K$ is

$$N(x) = \prod_{\sigma \in G} \sigma(x)$$

Lemma 1.2. The following statements are equivalent:

1. $(a, b)_p = 1$;
2. The quadratic equation $z^2 - ax^2 - by^2 = 0$ has a non-trivial solution;
3. $a \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{b})$.
4. $b \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{a})$.

Proof. For $2 \implies 3$, if $b \in \mathbf{Q}_p^{\times 2}$, then we are done. Otherwise, □

For $p \neq 2, \infty$ and $a, b \in \mathbf{Z}$, write $a = p^{v_p(a)}u$ and $b = p^{v_p(b)}v$ with $p \nmid uv$. The relation between the Hilbert symbol and the Legendre symbol is

$$(a, b)_p = (-1)^{\frac{p-1}{2}v_p(a)v_p(b)} \left(\frac{u}{p}\right)^{v_p(b)} \left(\frac{v}{p}\right)^{v_p(a)} \quad (1.1)$$

$$= \overline{(-1)^{v_p(a)v_p(b)} a^{v_p(b)} b^{-v_p(a)}}^{(p-1)/2}. \quad (1.2)$$

where the bar denotes reduction modulo p . In particular, if p does not divide a , then $(a, b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$ coincides with the Legendre symbol.

Theorem 1.3 (Hilbert reciprocity law). For all $a, b \in \mathbf{Q}^\times$, we have

$$\prod_p (a, b)_p = 1$$

As for computation, one shows that if p does not divide $2ab$ (including $p = \infty$ for $a, b > 0$), then $(a, b)_p = 1$. Moreover, $(a, b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}$ for odd a, b . From this and multiplicativity in the first variable, we immediately deduce the quadratic reciprocity law by taking a, b to be odd primes.

An advantage of this formulation is that it generalizes to number fields K , by considering the completions of K at all of its (finite and infinite) “places”. Hilbert then went on to pose his 9th problem, in which he asked for a generalization of this law to n -th powers in arbitrary number fields; this is what class field theory achieves. We first consider the case of extensions which contain a primitive n -th root of unity:

1.2 Kummer case

Assume now $K|\mathbf{Q}$ is a finite extension such that $\mu_n \subset K$, and let $\mathcal{O}_K$ be its ring of integers. If $P \subset \mathcal{O}_K$ is a prime ideal, we would like to define a symbol $(a, b)_P \in \mu_n$ such that a product formula holds, i.e.

$$\prod_P (a, b)_P = 1$$

if $a, b \in K^\times$. Moreover, for P finite (which is always the case for $n \neq 2$) which does not divide (n) we shall have

$$(a, b)_P = \overline{(-1)^{v_P(a)v_P(b)} a^{v_P(b)} b^{-v_P(a)} }^{(q-1)/n},$$

where $q = \#(\mathcal{O}_K/P)$. In particular, if P does not divide (a) , then $(a, b)_P = a^{\frac{q-1}{n} v_P(b)} \pmod{P}$.

So, if $v_P(b) = 1$, then

$$\left(\frac{a}{P}\right) := (a, b)_P$$

does not depend on the choice of b and it is the unique n -th root of unity, that is $a^{\frac{q-1}{n}} \pmod{P}$; in other words, $\left(\frac{a}{P}\right) = 1$ if and only if $x^n = a \pmod{P}$ has a solution. Some more properties of this symbol are:

1. $(a, b)_P = 1 \iff b$ is a norm from $K_P(\sqrt[n]{a}) \iff a$ is a norm from $K_P(\sqrt[n]{b})$;
2. $(a, b)_P$ is multiplicative in both variables and anticommutative;

When μ_n is *not* contained in K_P , an ‘‘Hilbert’’ symbol still exists but is longer associated to $a, b \in K^\times$. Suppose for a moment that $\mu_n \subset K_P$. Then, the extension $K_P(\sqrt[n]{a})|K_P$ is Galois for all $a \notin K_P^{\times n}$, and its Galois group is

$$G_a := \text{Gal}(K_P(\sqrt[n]{a})|K_P) \cong \mathbf{Z}/n\mathbf{Z}$$

Then, the group $\text{Hom}(G_a, \mu_n) \cong \mathbf{Z}/n\mathbf{Z}$ is generated by the character

$$\overline{\chi}_a : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n \cong \mathbf{Z}/n\mathbf{Z}.$$

In general, we shall construct a generalized Hilbert symbol

$$\begin{aligned} \text{Hom}(\text{Gal}(\overline{K_P}|K_P), \mathbf{Z}/n\mathbf{Z}) \times K_P^\times &\rightarrow \mathbf{Z}/n\mathbf{Z}, \\ [\chi, b] &\mapsto (\chi, b)_P \end{aligned}$$

and we’ll thus be able to prove the **Artin** reciprocity law: fix some $b \in K^\times$, and suppose that χ is a character in $\text{Hom}(G_K, \mathbf{Z}/n\mathbf{Z})$. Then

$$\sum_P (\chi_P, b) = 0,$$

where χ_P is the restriction of χ to $G_{K_P}^{\text{ab}}$ (recall that it embeds into G_K^{ab}).

The generalized Hilbert symbol defines a reciprocity map

$$K_P^\times \rightarrow G_{K_P}^{\text{ab}},$$

which is ‘‘almost’’ an isomorphism. In particular, for every n we’ll have an isomorphism of finite groups

$$K_P^\times/n \rightarrow G_{K_P}^{\text{ab}}/n.$$

To define such a map, it’s enough to define the dual map

$$\begin{aligned} \text{Hom}(G_{K_P}^{\text{ab}}/n, \mathbf{Z}/n\mathbf{Z}) &\rightarrow \text{Hom}(K_P^\times/n, \mathbf{Z}/n\mathbf{Z}), \\ [\chi] &\mapsto [b \mapsto (\chi, b)_P]. \end{aligned}$$

References

- [1] James S. Milne, ...